

Diagnosing Hardware Failure Theory

1. List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' — symptom: blank screen, no POST beep).

5 Common Desktop PC Hardware Issues:-

Hardware Issue	Typical Symptom
RAM Failure	PC may not boot, display may remain blank, or repeated beep sounds may occur.
Hard Disk/SSD Failure	Slow system, boot errors, freezing, or “Operating System not found” message.
SMPS/Power Supply Failure	PC does not turn on, sudden shutdown, or no power to components.
Overheating	PC becomes very hot, slows down, freezes, or shuts down automatically.
Graphics Card Failure	No display, screen artifacts, flickering, or crashes during graphics-intensive tasks.

2. Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.

Laptop Not Charging – Step-by-Step Troubleshooting:-

Step 1: Check the power socket and make sure electricity is available.

Step 2: Check the charger and cable for any cuts, loose connections, or physical damage.

Step 3: Check the laptop charging port and remove any dust or dirt carefully.

Step 4: Plug the charger firmly into the laptop and check whether the charging indicator light turns on.

Step 5: Try another compatible charger to find out whether the problem is with the charger.

Step 6: If available, use a **multimeter** to check whether the charger is giving the correct output voltage.

Step 7: Check the battery for swelling, damage, or loose connection.

Step 8: If the charger and battery are working properly, the **charging port or motherboard charging circuit** may be faulty and should be checked by a technician.

3. Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).

USB Charger Voltage Test Using a Multi meter :-

Step 1: Set the multimeter to **DC Voltage (V_{DC})** mode.

Step 2: Connect the **black probe to GND** and the **red probe to the +5V pin** of the USB connector.

Step 3: Turn on/connect the USB charger to the power supply.

Step 4: Check the voltage displayed on the multi meter.

Step 5: Record the reading. For example: **5.02 V**.

Step 6: Compare the reading with the expected USB range of **4.75 V to 5.25 V**.

4. A PC is randomly shutting down. Use a multi meter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate?

Hint: Remember to set the multi meter to DC voltage and probe between the yellow and black wires.

Testing the ATX 12V Rail Using a Multi meter:-

Step 1: Switch off the PC and **disconnect the SMPS from the mains**.

Step 2: Disconnect the ATX power connector from the motherboard.

Step 3: Set the multimeter to **DC Voltage (V_{DC})**.

Step 4: Connect the **black probe to a black wire (GND)** and the **red probe to the yellow wire (+12V)**.

Step 5: Turn on the power supply only if you know the ATX testing procedure and can do so safely.

Step 6: Record the voltage shown on the multimeter.